Finite generation of pluricanonical rings

Christopher Hacon

University of Utah

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Outline of the talk

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Notation

- I am interested in the birational classification of complex projective varieties.
- $X \subset \mathbb{P}^N_{\mathbb{C}}$ is defined by homogeneous polynomials $P_1, \cdots, P_t \in \mathbb{C}[x_0, \cdots, x_N]$.
- X, X' are birational if they have isomorphic open subsets $U \cong U'$.
- If X is smooth of dimension n, then T_X denotes the **tangent bundle** of X and $\omega_X = \Lambda^n T_X^\vee$ is the **canonical line bundle** of X.
- K_X denotes (a choice of) the canonical divisor i.e. a formal linear combination of codimension 1 subvarieties of X given by the zeroes of a section $s \neq 0$ of ω_X.
- We would like to use ω_X to understand the geometry of X.

Birational invariants

- If X and X' are birational, then $\forall m \geq 0$
$$H^0(X, \omega_X^{\otimes m}) \cong H^0(X', \omega_{X'}^{\otimes m}).$$
- The numbers $P_m(X) := h^0(\omega_X^{\otimes m})$ are the **plurigenera** of X.
- The **pluricanonical ring** of X is given by
$$\mathfrak{R}(K_X) = \bigoplus_{m \in \mathbb{N}} H^0(X, \omega_X^{\otimes m}).$$
- The **Kodaira dimension** of X is
$$\kappa(X) := \mathrm{tr.deg.}_{\mathbb{C}} \mathfrak{R}(K_X) - 1.$$
- If $\kappa(X) = \mathrm{dim} X$, we say that X is of **general type**.
- One would like to understand the structure of this ring and to use its features to classify complex projective varieties.

Finite generation

Today I would like to discuss a recent result of Birkar-Cascini-Hacon-M^{c}Kernan and Siu:

Theorem

Let X be a smooth projective variety, then $\mathfrak{R}(K_X)$ is finitely generated.

The two proofs are independent. Our proof is algebraic and uses the ideas of the MMP (minimal model program). Siu's proof is analytic and requires X to be of general type.

Curves

- Any two birational curves are isomorphic.
- Curves are topologically determined by their genus.
- $g = 0$: $\mathbb{P}^1$ and $\mathfrak{R}(K) \cong \mathbb{C}$
- $g = 1$: There is a 1-dimensional family of elliptic curves; $K = 0$ so $\mathfrak{R}(K) \cong \mathbb{C}[t]$.
- $g \geq 2$: Curves of general type.
  These form a $3g - 3$-dimensional family.
  K is ample and $|3K|$ determines an embedding.
- $\mathfrak{R}(K)$ is then finitely generated (by Serre-Vanishing).

Surfaces of general type

- If S is a surface of general type and K_S is nef (i.e. $K_S \cdot C \geq 0$ for all curves $C \subset S$), then we say that S is a minimal surface. In this case K_S is semiample.
- Therefore, there is a birational morphism $\nu : S \to \bar{S} \subset \mathbb{P}^N$ to a surface (with rational double points) so that $\omega_X^{\otimes m} = \nu^* \mathcal{O}_{\bar{S}}(1)$.
- It is known that $\mathfrak{R}(K_S)$ is finitely generated if and only if
$$\mathfrak{R}(K_S)^{(m)} = \bigoplus_{i \in \mathbb{N}} H^0(\mathcal{O}_S(imK_S))$$
is finitely generated (for any $m > 0$).
- By Serre vanishing $\mathfrak{R}(K_S)^{(m)} \cong \mathfrak{R}(\mathcal{O}_{\bar{S}}(1))$ is finitely generated.
- Note that we may take $m = 5$, ν contracts rational curves E with $E \cdot K_S = 0$ and $\bar{S} = \mathrm{Proj}\mathfrak{R}(K_S)$.

Surfaces of general type II

- If S is of general type, but K_S is not nef, then there is a rational curve E such that $E^2 = E \cdot K_S = -1$.
- We may contract E, i.e. there is a morphism of smooth surfaces $\nu_1 : S \to S_1$ which is an isomorphism on $S - E$ and such that E maps to a point.
- Since $K_S = \nu_1^* K_{S_1} + E$, we have $\mathfrak{R}(K_S) \cong \mathfrak{R}(K_{S_1})$.
- If K_{S_1} is not nef, we may repeat the procedure and we obtain a sequence of morphisms $S \to S_1 \to S_2 \to \cdots$.
- It is easy to see that $0 < \rho(S_{i+1}) = \rho(S_i) - 1$ and so this sequence of morphism is finite, i.e. there is an integer $t > 0$ such that K_{S_t} is nef i.e. S_t is a minimal model of S.
- Then $\mathfrak{R}(K_{S_t})$ is finitely generated and so is $\mathfrak{R}(K_S)$.

Surfaces with $\kappa \leq 2$

- If S is not of general type, then $\kappa(S) \in \{-1, 0, 1\}$.
- If $\kappa(S) = -1$, then $\mathfrak{R}(K_S) \cong \mathbb{C}$.
- If $\kappa(S) = 0$, then $\mathfrak{R}(K_S)^{(m)} \cong \mathbb{C}[t]$ for some $m > 0$. (For surfaces $m = 12$.)
- If $\kappa(S) = 1$, then we first contract all -1 curves to obtain a morphism $\nu : S \to S'$ such that $K_{S'}$ is nef (i.e. S' is minimal). Then $K_{S'}$ is semiample and it gives a morphism $f : S' \to C$ whose general fiber is an elliptic curve.
- We have that $mK_{S'} = f^*H$ where $m > 0$ is an integer and H is an ample divisor on C.
- Therefore, $\mathfrak{R}(K_S)^{(m)} \cong \mathfrak{R}(K_{S'})^{(m)} \cong \mathfrak{R}(H)$ is finitely generated.

Outline of the talk

Log pairs

- One would like to generalize the above approach to higher dimensional varieties.
- In order to achieve this one has to allow varieties with mild singularities and the minimal models will not be unique.
- We would also like to have more flexibility and so we work with log-pairs.
- The proof will in fact proceed by induction on the dimension. We will need to relate K_X to it's restriction to a divisor S (which typically belongs to $S \in |mK_X|$).
- This is achieved by the adjunction formula which has the form $(K_X + S)|_S = K_S + \Delta_S$ for some $\mathbb{Q}$-divisor Δ_S.

KLT Log Pairs

- We consider log pairs (X, Δ) where X is a normal $\mathbb{Q}$-factorial variety and Δ is a divisor with positive real coefficients.
- Then $K_X + \Delta$ is $\mathbb{R}$-Cartier and so $(K_X + \Delta) \cdot C$ and $f^*(K_X + \Delta)$ still make sense.
- Consider a map $f : X' \to X$ such that X' is smooth and $f^*\Delta + \mathrm{Exc}(f)$ has simple normal crossings support. Write

$$K_{X'} + \Delta' = f^*(K_X + \Delta).$$

- (X, Δ) is KLT if each coefficient of Δ' is less than 1.

A Theorem of Fujino-Mori

Mori and Fujino have shown that:

Theorem

Let (X, Δ) be any KLT $\mathbb{Q}$-factorial pair with $\Delta \in \mathrm{Div}_{\mathbb{Q}}(X)$. Then there exists a **big** *KLT $\mathbb{Q}$-factorial pair (Y, Γ) with $\Gamma \in \mathrm{Div}_{\mathbb{Q}}(Y)$ such that*

$$\mathfrak{R}(K_X + \Delta)^{(m)} \cong \mathfrak{R}(K_Y + \Gamma)^{(m)}$$

for any sufficiently divisible integer $m > 0$.

Recall that $K_Y + \Gamma$ is big if $\mathrm{tr.deg.}_{\mathbb{C}}\mathfrak{R}(K_Y + \Gamma)^{(m)} = \dim Y$ or equivalently if $K_Y + \Gamma \sim_{\mathbb{Q}} A + B$ with A ample and $B \geq 0$.
It follows that to show the main theorem (i.e. that $\mathfrak{R}(K_X)$ is finitely generated), it suffices to prove the corresponding statement for big KLT log pairs.

The MMP with scaling

- From now on we will consider $\mathbb{Q}$-factorial KLT pairs (X, Δ) such that Δ is big. We allow $\Delta \in \mathrm{Div}_{\mathbb{R}}(X)$.
- Note that if $K_X + \Delta$ is big, then there is an effective divisor $D \sim_{\mathbb{R}} K_X + \Delta$ and so for all $0 < \epsilon \ll 1$ the pair $(X, \Delta + \epsilon D)$ is KLT, $\Delta + \epsilon D$ is big and $K_X + \Delta + \epsilon D \sim_{\mathbb{R}} (1+\epsilon)(K_X + \Delta)$.
- We would like to find a finite sequence of KLT log pairs (X_i, Δ_i) with Δ_i big and birational maps $X \dashrightarrow X_1 \dashrightarrow X_2$ such that

$$\mathfrak{R}(K_{X_i} + \Delta_i) \cong \mathfrak{R}(K_{X_{i+1}} + \Delta_{i+1}),$$

  and $K_{X_t} + \Delta_t$ is nef.
- Then by the base point free theorem, $K_{X_t} + \Delta_t$ is semiample so that $\mathfrak{R}(K_{X_t} + \Delta_t)$ is finitely generated (if $\Delta \in \mathrm{Div}_{\mathbb{Q}}$).

The MMP with scaling II

- Assume that A is ample and $K_X + \Delta + A$ is nef.
- Let $\lambda = \inf\{t \geq 0 | K_X + \Delta + tA\}$ is nef.
- If $\lambda = 0$ we STOP ($K_X + \Delta$ is nef).
- Otherwise, by the cone theorem, there is an extremal ray R such that $(K_X + \Delta + \lambda A) \cdot R = 0$ and $(K_X + \Delta + tA) \cdot R < 0$ for $0 \leq t < \lambda$.
- Let $\mathrm{cont}_R : X \to Z$ be the corresponding morphism which contracts a curve C iff $[C] = R$.
- If cont_R is not birational, we STOP (we have a Mori-Fano fiber space and $\mathfrak{R}(K_X + \Delta) \cong \mathbb{C}$).

The MMP with scaling III

- If cont_R is birational and divisorial (i.e. it contracts a divisor), then we replace X by the image of the corresponding contraction and we may continue this procedure.
- If cont_R is birational and small, (i.e. $\mathrm{Exc}(\mathrm{cont}_R)$ has codimension ≥ 2) then the image of the corresponding contraction is not KLT. Therefore, we can NOT replace X by the corresponding contraction!
- Instead we replace (X, Δ) by the corresponding flip $\phi : X \dashrightarrow X^+$ (assuming it exists).
- Let $\Delta^+ = \phi_*\Delta$. The flip is a codimension 2 surgery which replaces $K_X + \Delta$ negative curves by $K_{X^+} + \Delta^+$ positive curves.

Flips

- More precisely, $f = \mathrm{cont}_R : X \to Z$ is a flipping contraction i.e. a small birational morphism with $\rho(X/Z) = 1$ and $-(K_X + \Delta)$ ample over Z.
- The flip $f^+ : X^+ \to Z$ (if it exists) is a small birational morphism such that $\rho(X^+/Z) = 1$ and $K_{X^+} + \Delta^+$ is ample over Z.
- The flip $f^+ : X^+ \to Z$ (if it exists) is given by

$$X^+ = \mathrm{Proj}_Z \bigoplus_{m \in \mathbb{N}} f_* \mathcal{O}_X(\lfloor m(K_X + \Delta) \rfloor).$$

- Note that after peturbing Δ, we may assume that $\Delta \in \mathrm{Div}_{\mathbb{Q}}(X)$.

Flips II

- The problem of existence of flips is local over Z and hence we may assume that Z is affine. It suffices then to show that $\mathfrak{R}(K_X + \Delta)$ is finitely generated. So this is a local version of the original problem.
- To run this MMP with scaling, we must therefore show that such flips exist and terminate (i.e. there is no infinite sequence of such flips).
- Note that after each divisorial contraction, the Picard number drops by 1 and so there are only finitely many divisorial contractions.

Outline of the talk

Existence and termination of flips

Theorem (Birkar-Cascini-Hacon-McKernan)

Let (X, Δ) be a KLT log pair and $f : X \to Z$ a flipping contraction, then the flip $f^+ : X^+ \to Z$ of f exists. If moreover Δ is big, then any sequence of flips for the $K_X + \Delta$ MMP with scaling must terminate.

Corollary

Let (X, Δ) be a KLT pair with $K_X + \Delta$-pseudo-effective and Δ big, then $K_X + \Delta$ has a minimal model. (In particular flips exist.)

$K_X + \Delta$-pseudo-effective means that given any ample $\mathbb{Q}$-divisor H, $K_X + \Delta + H$ is big (i.e. $\kappa(K_X + \Delta + H) = \dim X$). This is a necessary condition.

Finite Generation

Corollary (Birkar-Cascini-Hacon-McKernan, Siu)

Let X be a smooth projective variety, then $\mathfrak{R}(K_X)$ is finitely generated.

- Proof: Assume that X is of general type. We may find $\Delta \sim \epsilon K_X$ with $0 < \epsilon \ll 1$. A minimal model for $K_X + \Delta$ is then a minimal model for K_X. The result then follows from the base point free theorem.
- As mentioned above, the general case is a consequence of a result of Fujino and Mori.
- Siu's proof relies on analytic techniques.

The structure of the proof

We proceed by induction on $n = \dim X$. We show that for an n-dimensional KLT pair (X, Δ):

Claim 1. Flips exist.

Claim 2. If $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$ and Δ is big, then (X, Δ) has a minimal model.

Claim 3. The set of minimal models for KLT pairs with $A \leq \Delta \leq \Delta_0$ where A is ample and (X, Δ_0) is KLT, is finite.

Claim 4. If Δ is big and $K_X + \Delta$ is pseudo-effective, then there is a $\mathbb{Q}$-divisor D such that $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$.

All statements should be relative over a base.

The structure of the proof II

- Roughly speaking, the inductive structure of the proof is:

$$(1-4)_{n-1} \Rightarrow 1_n \Rightarrow 2_n \Rightarrow 3_n \Rightarrow 4_n.$$

- Note that that $(1-4)_{n-1} \Rightarrow 1_n$ was obtained by Hacon M$^{\mathrm{c}}$Kernan in 2005 using ideas of Shokurov and an extension result base on work of Siu, Kawamata and Tsuji.
- The remaining implications heavily use the ideas of the MMP established by Kawamata, Kollár, Mori, Reid, Shokurov and others.

Reduction to PL Flips

- Shokurov has shown that to construct flips it suffices to construct PL-flips.
- That is, we assume that Δ has a component S of multiplicity 1, $-S$ is relatively ample and $K_X + \Delta$ is "almost KLT".
- By adjunction $(K_X + \Delta)|_S = K_S + \Delta_S$ and (S, Δ_S) is KLT.
- As we have remarked above, the existence of flips is equivalent to showing that $\mathfrak{R} = \mathfrak{R}(K_X + \Delta)$ is finitely generated. (Here we assume Z is affine.)
- This is equivalent to showing that the algebra $\mathfrak{R}_S$ given by the image of the restriction map

$$\bigoplus_{m \in \mathbb{N}} H^0(\mathcal{O}_X(m(K_X + \Delta))) \to \bigoplus_{m \in \mathbb{N}} H^0(\mathcal{O}_S(m(K_S + \Delta_S)))$$

is finitely generated.

Existence of flips

- If $\mathfrak{R}_S = \oplus_{m\in\mathbb{N}} H^0(\mathcal{O}_S(m(K_S+\Delta_S)))$ we would be done by induction on the dimension.
- This is too much to hope for.
- Instead, we show that $\mathfrak{R}_S$ has very special properties.
- Roughly speaking we show that there exists a divisor $0 \leq \Theta \leq \Delta_S$ on S such that

$$\mathfrak{R}_S = \bigoplus_{m\in\mathbb{N}} H^0(\mathcal{O}_S(m(K_S+\Theta))).$$

- This is proved using some extension theorems that were inspired by work of Siu, Kawamata and Tsuji.
- These extension theorems determine/define Θ.

Termination of MMP with scaling

- Since we are unable to show that an arbitrary sequence of flips must terminate, we run a MMP with scaling.
- The idea is that if A is sufficiently ample, then $K_X + \Delta + A$ is ample and so it is a minimal model. We then consider $K_X + \Delta + tA$ and let t go to 0.
- Proceeding this way we get a sequence of pairs $(X_i, \Delta_i + \lambda_i A_i)$ such that $K_{X_i} + \Delta_i + \lambda_i A_i$ is nef and trivial on some $K_{X_i} + \Delta_i$ extremal ray. We then flip/contract.
- Each $(X_i, \Delta_i + \lambda_i A_i)$ is a minimal model for $(X, \Delta + \lambda_i A)$.

Termination of MMP with scaling II

- If we could assume Claim 3_n, we would be done. (As Δ is big, we may assume that $A' \leq \Delta \leq \Delta + tA$ for some ample divisor A' and we let $\Delta_0 = \Delta + A$.)
- Since we can only assume 3_{n-1}, we replace (X, Δ) by an appropriate DLT pair (X', Δ') and we run a MMP with scaling such that all flips/contractions are contained in $[\Delta']$.
- This step is similar to Shokurov's reduction to PL-flips.

Finiteness of models

Theorem

Assume that (X, Δ_0) is KLT, A is an ample $\mathbb{Q}$-divisor and $\Delta = \sum_{i=1}^r \delta_i \Delta_i$. Let $\mathcal{C} \subset [0,1]^r$ be a rational polytope such that for any $(\delta_1, \cdots, \delta_r) \in \mathcal{C}$, the pair $(X, \sum_{i=1}^r \delta_i \Delta_i + A)$ is KLT. Then the set of isomorphism classes

$$\{Y | Y \text{ is a min. model for } (X, \Delta + A);\ (\delta_1, \cdots, \delta_r) \in \mathcal{C}\}$$

is finite.

The proof is based on ideas of Shokurov (1996). It requires a compactness argument and so it is necessary to work with $\mathbb{R}$-divisors.

Finiteness of models II

- We pick $\Delta \in \mathcal{C}$ and work in a sufficiently small neighborhood U of Δ.
- After running a MMP with scaling, we may assume that $K_X + \Delta$ is nef.
- By the base-point free theorem, there is a morphism $f : X \to Z$ such that $K_X + \Delta \sim_{\mathbb{R}} f^*H$.
- Working over Z, we may assume that $K_X + \Delta \sim_{\mathbb{R}} 0$.
- Therefore $\mathcal{C} \cap U$ is a cone over Δ and we are done by induction on the dimension of the subspace of $\mathrm{Div}_{\mathbb{R}}(X)$ spanned by the components of Δ.

Non-vanishing

- We must show that if Δ is big and $K_X+\Delta$ is pseudo-effective, then $K_X+\Delta \sim_{\mathbb{R}} D \geq 0$.
- If $h^0(\mathcal{O}_X(m(K_X+\Delta)+A))$ is bounded (for A ample), then the result follows from work of Nakayama.
- If $h^0(\mathcal{O}_X(m(K_X+\Delta)+A))$ is un-bounded, then we can produce a log canonical center not contained in the stable base locus of $K_X+\Delta+\epsilon A$ for $0<\epsilon\ll 1$.
- So we study a log smooth PLT pair (X',Δ') where $S=[\Delta']$ is irreducible.
- Then $K_S+(\Delta'-S)|_S$ is pseudo-effective, $(\Delta'-S)|_S$ is big and so $h^0(\mathcal{O}_S(m(K_S+(\Delta'-S)|_S)))>0$ for some $m>0$.
- Using Kawamata-Viehweg vanishing, we can lift this section to $m(K_X+\Delta)$.

Open problems

- Abundance: If $K_X + \Delta$ is nef and KLT show that it is semiample. (When Δ is big, this follows from the Base Point Free Theorem. In dim. 2, it follows from the classification. It also holds in dim. 3)
- Existence of Minimal models for Log Canonical pairs (should be equivalent to Abundance).
- Termination of flips (would follow from the Borisov-Alexeev-Borisov conjecture).
- Characteristic p (need resolution of singularities; there is no Kawamata-Viehweg vanishing Thm, but ....).